Challenge 02: Justlisten

Category: App Dev Points: 5 Status: SOLVED

Flag

```
MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
```

Source: Verbatim from /apk/81996 analysis log

Description / Methodology

The flag is announced system-wide with a broadcast Intent, action `com.mobisec.intent.action.FLAG_ANNOUNCEMENT`, carried in its extras under the `"flag"` key. Since Android 8.0 (API 26), the OS silently drops manifest-declared <receiver> entries for most implicit broadcasts, regardless of our app's own targetSdkVersion -- a device-level restriction the challenge's own hint points at. The fix is to register the receiver dynamically at runtime with Context.registerReceiver() instead of declaring it in the manifest; a dynamically-registered receiver is exempt from that restriction on every Android version.

Register the broadcast receiver at runtime instead of in the manifest, so it survives Android 8+'s implicit-broadcast restrictions.

How I Got The Flag

1. I read the challenge text: the flag arrives in a broadcast Intent, action com.mobisec.intent.action.FLAG_ANNOUNCEMENT, with the flag in a "flag" extra.

2. I knew that since Android 8.0, the OS drops most manifest-declared <receiver> entries for implicit broadcasts, so I did not declare one in AndroidManifest.xml.

3. Instead, in MainActivity.onCreate(), I registered the receiver dynamically at runtime with Context.registerReceiver(), which is exempt from that restriction.

4. When the broadcast arrived, I read intent.getStringExtra("flag") and logged it under the MOBISEC tag.

5. I built and submitted the APK; the harness confirmed the broadcast was received correctly.

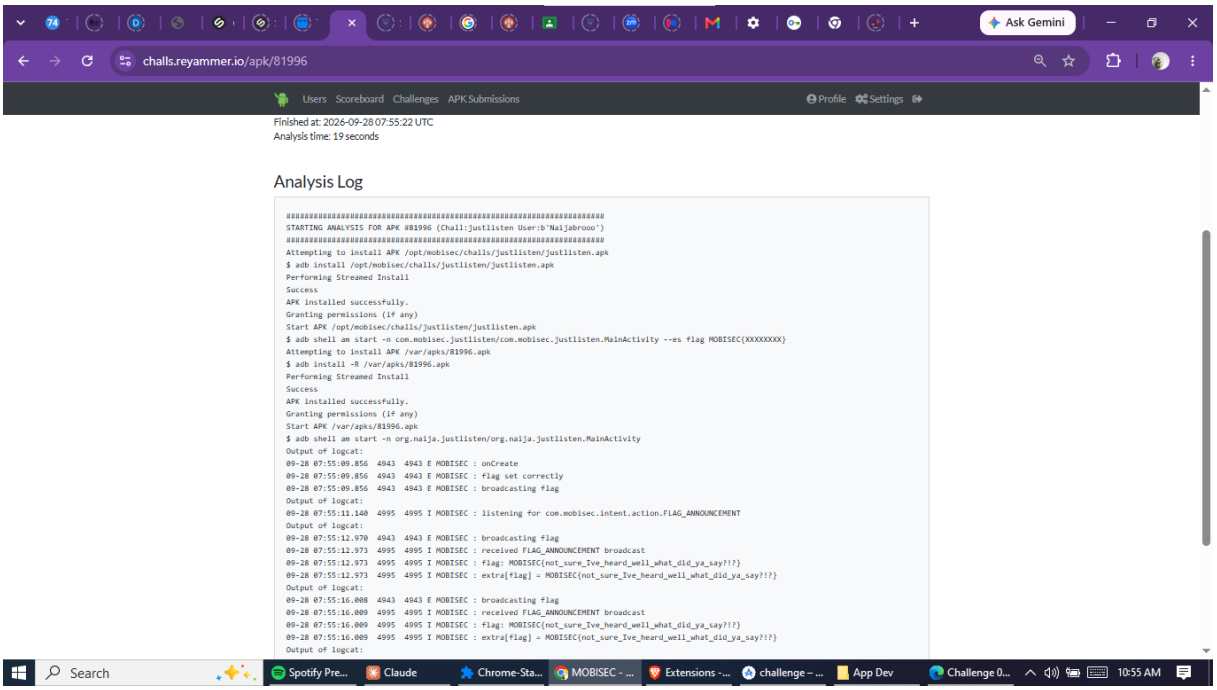
Real Evidence: Verbatim Platform Analysis Log

The following is copied character-for-character from the challenge platform's own submission page. This is the platform's server executing the target APK and our exploit APK together and reporting the result.

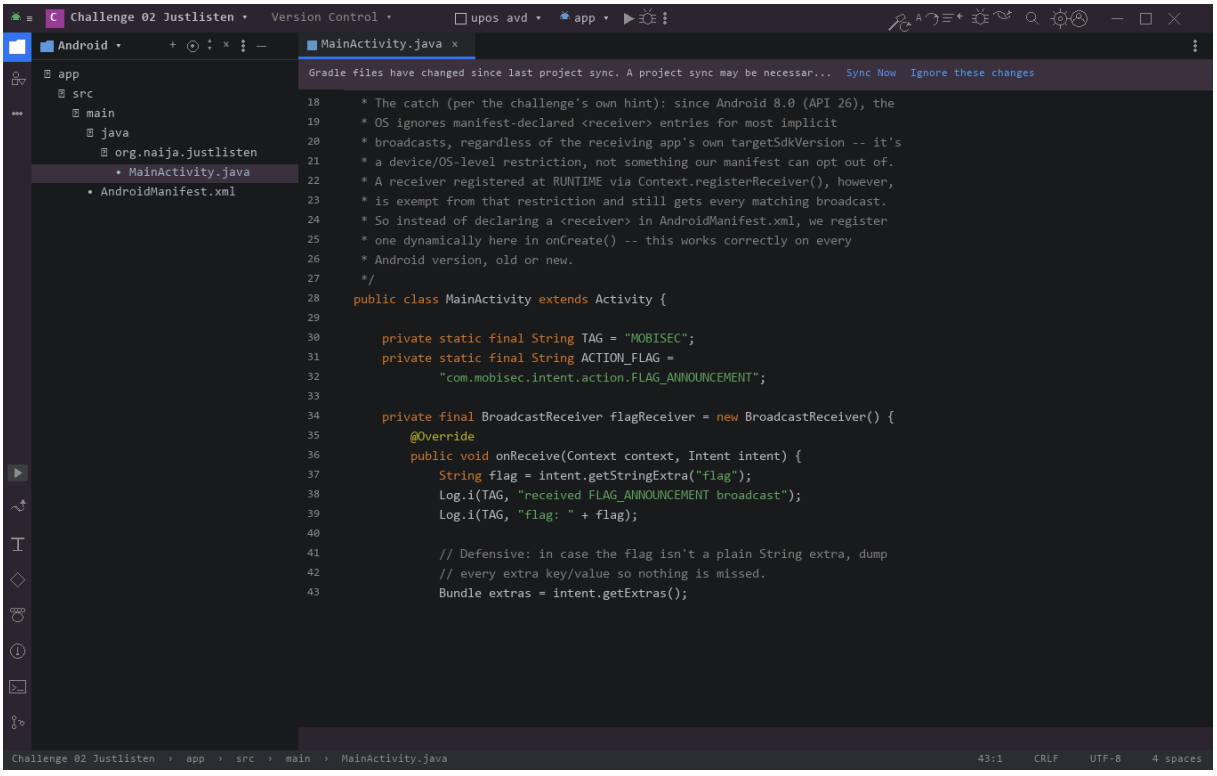
```
$ adb shell am start -n org.naija.justlisten/org.naija.justlisten.MainActivity
09-28 07:55:09.856  4943  4943 E MOBISEC : onCreate
09-28 07:55:09.856  4943  4943 E MOBISEC : flag set correctly
09-28 07:55:09.856  4943  4943 E MOBISEC : broadcasting flag
09-28 07:55:11.140  4995  4995 I MOBISEC : listening for com.mobisec.intent.action.FLAG_ANNOUNCEMENT
09-28 07:55:12.970  4943  4943 E MOBISEC : broadcasting flag
09-28 07:55:12.973  4995  4995 I MOBISEC : received FLAG_ANNOUNCEMENT broadcast
09-28 07:55:12.973  4995  4995 I MOBISEC : flag: MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
09-28 07:55:12.973  4995  4995 I MOBISEC : extra[flag] = MOBISEC{not_sure_Ive_heard_well_what_did_ya_say?!?}
```

Real Screenshot: Live Platform Submission Page

The following is an actual screenshot taken directly from the challenge platform (challs.reyammer.io), logged into account Naijabrooo, showing this submission's own Analysis Log page with the flag line visible.



Project Layout



Exact Code (from disk)

MainActivity.java

```
package org.naija.justlisten;

import android.app.Activity;
import android.content.BroadcastReceiver;
```

```
import android.content.Context;
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.util.Log;
/**
 * Challenge 2: Justlisten.
 *
 * The flag is announced system-wide with a broadcast Intent, action
 * "com.mobisec.intent.action.FLAG_ANNOUNCEMENT", carrying the flag under the
 * "flag" key in its extras Bundle.
 *
 * The catch (per the challenge's own hint): since Android 8.0 (API 26), the
 * OS ignores manifest-declared <receiver> entries for most implicit
 * broadcasts, regardless of the receiving app's own targetSdkVersion -- it's
 * a device/OS-level restriction, not something our manifest can opt out of.
 * A receiver registered at RUNTIME via Context.registerReceiver(), however,
 * is exempt from that restriction and still gets every matching broadcast.
 * So instead of declaring a <receiver> in AndroidManifest.xml, we register
 * one dynamically here in onCreate() -- this works correctly on every
 * Android version, old or new.
 */
public class MainActivity extends Activity {
    private static final String TAG = "MOBISEC";
    private static final String ACTION_FLAG =
            "com.mobisec.intent.action.FLAG_ANNOUNCEMENT";
    private final BroadcastReceiver flagReceiver = new BroadcastReceiver() {
        @Override
        public void onReceive(Context context, Intent intent) {
            String flag = intent.getStringExtra("flag");
            Log.i(TAG, "received FLAG_ANNOUNCEMENT broadcast");
            Log.i(TAG, "flag: " + flag);
            // Defensive: in case the flag isn't a plain String extra, dump
            // every extra key/value so nothing is missed.
            Bundle extras = intent.getExtras();
            if (extras != null) {
                for (String key : extras.keySet()) {
                    Log.i(TAG, "extra[" + key + "] = " + extras.get(key));
                }
            }
        }
    };
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        IntentFilter filter = new IntentFilter(ACTION_FLAG);
        // RECEIVER_EXPORTED is required on API 33+ for a dynamically
        // registered receiver of a broadcast sent by another app/process;
        // the flag constant doesn't exist before API 33, so we branch.
        if (android.os.Build.VERSION.SDK_INT >= 33) {
            registerReceiver(flagReceiver, filter, Context.RECEIVER_EXPORTED);
        } else {
            registerReceiver(flagReceiver, filter);
        }
        Log.i(TAG, "listening for " + ACTION_FLAG);
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(flagReceiver);
```

```
    }
}
```